

Azure SQL DB Competitive Analysis

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Purpose/Objective

This document focuses solely on Azure SQL Database offerings, our competitive messaging and feature comparison to provide comprehensive guidance during Architecture and Technical Sales conversation to better position RDS for SQL Server. We expect this document will cater 70% of use cases and Azure SQL related compete scenario including RFP/RFI inputs. Also this document comprises of feature gaps which RDS SQL Server currently have with Azure SQL offerings which will provide helpful insights to our service team to gauge the impact and invest accordingly. For any further insights or questions please engage Service Specialist SA through SpecReq. Please note performance benchmarking between Azure SQL Services & Amazon RDS for SQL Server will be a separate document and currently not in scope of this paper.

Executive Summary

Microsoft SQL Server is one of the most commonly used relational database engines (the specific software that handles data storage, organization, queries, and more). Although Open Source options such as PostgreSQL and MySQL are rising in popularity, many existing workloads are reliant on SQL Server and project teams continue to choose it on new applications for a variety of reasons including familiarity, historic popularity, and guaranteed support from Microsoft. The Azure SQL family is currently made up of four different services (Azure SQL Database/Elastic Pool, Azure SQL Managed Instance, SQL Server on Azure Virtual Machines, Azure SQL Edge). This document focuses on the managed database options, specifically Microsoft Azure SQL Database and Managed Instance, compared with Amazon Relational Database Service (RDS) for SQL Server. Amazon RDS is a compelling option that can cover the breadth of configurations offered by Azure SQL. Additionally, RDS offers the security and resilience of AWS and integration with a breadth of services providing the greatest depth in Analytics, AI, Machine Learning, Compute, and more. Below are few key findings across Migration, Licensing, HA, DR, Backup/Restore & Storage/Compute when comparing with Azure SQL Managed services offering.

1) Migration

- SQL Managed Instance databases can only be restored to SQL Server 2022 (either on-premises, or on a virtual machine) if the source SQL Managed Instance has enrolled in the November 2022 feature wave.
- SQL Managed Instance databases are encrypted with TDE by default. To restore your database to an instance other than the source SQL Managed Instance, the target instance must have access to the same key in Azure Key Vault used to encrypt the

source database, or you must disable TDE encryption on the source database before taking the backup.

- Azure has a clear advantage with new feature known as 'Managed Instance link feature'. The Managed Instance link feature uses distributed availability groups to extend your SQL Server on-premises Always On availability group hosted anywhere to Azure SQL Managed Instance in a safe and secure manner, replicating data in near real-time. The link supports single node SQL Server instances without existing availability groups, and also multiple-node SQL Server instances with existing availability groups. Through the link, customer can use the latest benefits of Azure without migrating your entire SQL Server data estate to the cloud

2) Security

- Overall, both AWS & Azure provides security strategies to protect database with layered defense-in-depth approach providing networking security, access management and encryption service for data protection. Therefore, have very similar concepts of VPC, Security groups, Network ACL to protect the database environment. But Azure has made progress to help customers improve their overall security strategies and posture by providing security related features natively on Azure Platform. These capabilities address the needs of the customers for secure database access, implement threat detection and vulnerability scanning solution.
- Azure AD now extended that capabilities to support kerberos authentication for Windows and SQL Server authentication. Customers now don't need to manage Active directory on a Windows Server as they did for years on-premises to support kerberos authentication. Azure position this feature as an effort to modernize the security infrastructure.

3) Compute, Memory & Storage

- AWS offers several types of [instance classes](#) for I/O, Memory and general-purpose workloads with max size up to 16 TB (along with private preview program of 64 TB) with 100 databases per instance. RDS SQL Server provides higher CPU to memory ratio with X instance class compare to Azure SQL. Also, AWS customers can use Provisioned IOPS on RDS SQL where in Azure customer have to scale higher tier to get more IOPS & throughput which eventually increase the cost.
- Azure has significant advantage as it gives customer flexibility to host a single database and pay for just one database rather than for whole SQL instance. With Azure SQL Elastic Pool, customer have significant advantage to host upto 5000 databases (mostly for multitenant SAAS applications) in a single pool in a cost effective & secured manner. To host similar number of databases we can recommend customer to use Amazon RDS Custom for SQL Server .However Amazon RDS Custom for SQL Server being a partially managed service is outsmarted by

managed features provided by Azure SQL Elastic Pool plus will be expensive solution to recommend to customer with License Included Model .

Azure SQL Managed Instance announced public preview of [Next-gen General Purpose \(GP\) service tier](#) . Under the hood, the Next-gen GP is built using the latest technology from Azure Storage: premium managed disks. This upgrade enables major enhancements in the areas of storage performance, instance scalability and cost optimization. It is recommended to showcase RDS SQL Server can achieve sub millisecond latency with IO2 Block Express compare to Nex Gen Azure Managed Instance which provides 3-5 ms latency and upto 80,000 IOPS.

4) High Availability

- Azure SQL guarantees up to [99.99%](#) availability of service. It means service can be down only for 4 minutes in a month or 52 minutes in a year v/s Amazon RDS for SQL Server being 99.95% which can be down up to 21 minutes a month or 4 Hour 20 minutes in a year. Having said that majority of the database workload can be hosted on Amazon RDS for SQL Server.
- The premium/business-critical tier locally redundant availability model for Azure SQL database provides access to one internal secondary replica to offload the read workload using AOAG technology.
- For database replication, Azure SQL DB customers can use active-geo replication [active geo-replication](#) feature to create readable secondary database both in-region and cross-region using async replication. Applications can easily recover from the complete or partial loss of an Azure region due to a natural disaster, catastrophic human errors. Customers can also use active-geo replication for other use-cases such as to support database migration between Azure regions and [managing rolling application upgrades](#).
- Azure provides capability where the read-write and read-only listener **end-points remain unchanged** during geo-failovers using **Auto-Failover group feature**. After the geo-failover is completed, the DNS record is automatically updated to redirect the endpoints to the new region.

5) Backup & Recovery

- Azure SQL Managed Instance provides [PITR recovery](#) for individual User Databases. Customers can use the Azure portal, PowerShell, the Azure CLI, or the Azure SDK. PiTR uses the automatic Full, Differential, and Log backups.
- By default, Azure SQL Database stores backups in geo-redundant [storage blobs](#) that are replicated to a [paired region](#). Geo-redundancy allows customers to restore their databases in a different region in the event of a regional outage.

6) Licensing

- Lastly, licensing is a frequent tactic Microsoft uses to compete with RDS. License Mobility is not supported in Amazon RDS for SQL Server (RDS SQL Custom does support BYOM though) , and Microsoft has an Azure Hybrid benefit that allows customers to bring on-premise licenses to the cloud. Remind the customers that Azure Hybrid benefit requires Software Assurance (which is a significant extra cost they must pay every year, perpetually). It is also not the overall TCO of a SQL server, so try the tips in the recommendations section below to compete on price.

Current Landscape

Microsoft Azure is a cloud computing platform operated by Microsoft. Azure provides software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS) offerings through a global network of Microsoft-owned data centers. Windows Azure was announced in October 2008 and launched in 2010 as a PaaS offering for Windows and SQL server on the cloud, but the offering did not get traction. In 2014, Microsoft renamed it to Microsoft Azure and started offering IaaS services and supporting Linux. Since the current offering is most aligned -architecturally and commercially – with the offering launched in 2014, the industry often refers to Azure as being ~8 years old.

Microsoft’s global infrastructure consists of 48 [generally available](#) regions compared to AWS 99 Availability Zones within 31 geographic regions around the world. Out of these 48 regions, only 27 have three AZs. Microsoft often touts having more regions (65+); their claim includes regions announced but not yet launched, and ~10 “Reserved Access Regions” that are not available to all customers and support a subset of customer scenarios. Please see Appendix section for [Azure regions with availability zone support](#).

Microsoft is viewed as the [number 2](#) cloud service provider [behind AWS and ahead of Alibaba, Google, and others](#), according to market research firms including [Gartner](#) and [Synergy Research](#).

Microsoft Azure Databases Overview

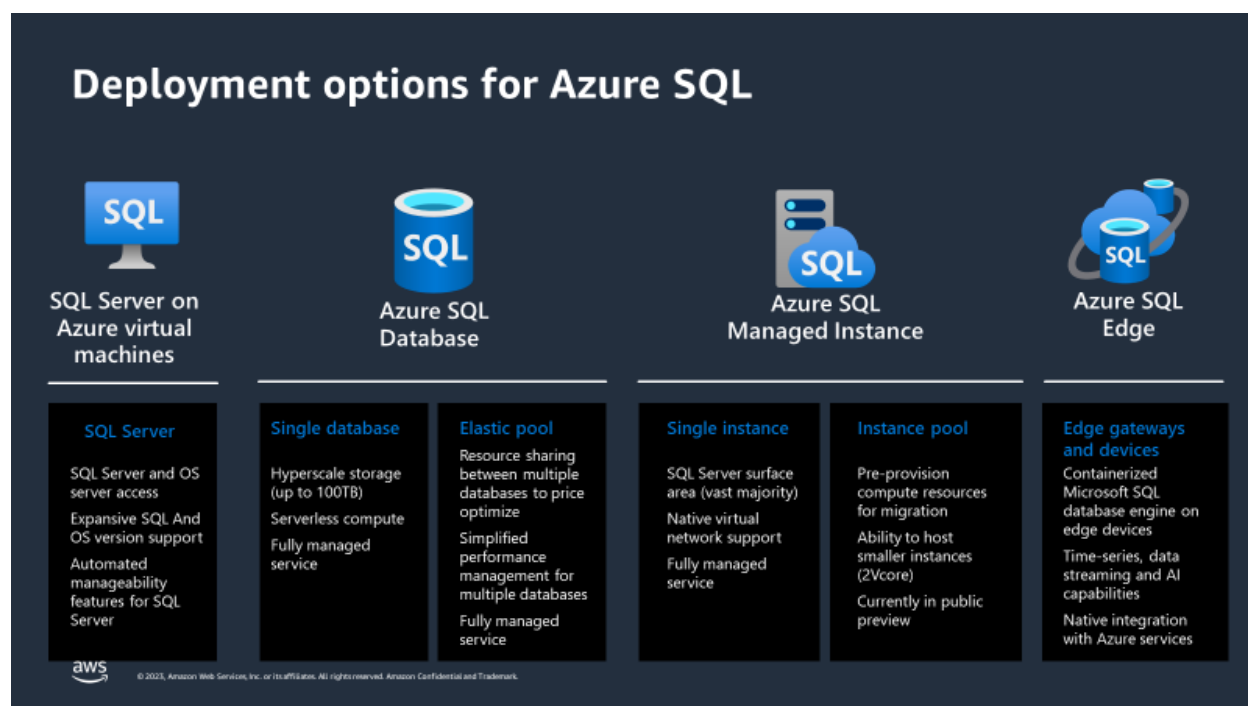
Microsoft Azure provides a broad portfolio of database services including relational, analytical, NoSQL, and in-memory databases. Much of Microsoft Azure success has been a result of Microsoft building on the success of SQL Server on-premises. Their wide variety of tools, skilled partner offerings, and large group of existing SQL users all have their roots in the popular on-prem database. Today, Microsoft is building on this momentum through their Azure SQL offerings and expanding their footprint to provide a variety of Open-Source and non-relational databases as well (3 RDS open source engines, 7 non-relational databases). When it comes to relational databases, Microsoft Azure offering is built around the SQL Server product. Azure provides customers the option to run SQL Server on a Virtual Machine or as a managed PaaS service. Azure SQL PaaS services include single database, Elastic pools or SQL Managed Instance. With single database offering, customers can create a cloud native database without the management overhead of a VM. Elastic Pools allow customers to share compute and storage resources across several databases, improving cost efficiency. Azure also offers serverless and Hyperscale (up to 100TB) features as part of Azure Single DB service. Lastly, SQL Managed Instance offers maximum compatibility with on-premises SQL Server for customers who are planning to migrate their legacy application to the Cloud. Amazon RDS for SQL Server is often compared with Azure SQL Managed Instance however RDS SQL Server is more preferred for Cloud Native applications and we have covered that aspect in more detail in this whitepaper.

Azure has also added support for Open Source databases like MySQL, PostgreSQL and MariaDB, CassandraDB, and MongoDB. Customers who are looking to leverage Open Source offerings can leverage Azure SQL Database for MySQL, PostgreSQL, or MariaDB as well as CosmosDB's support for Cassandra or MongoDB bindings. Microsoft's acquisition of Citus Data in 2019, resulted in Postgres Hyperscale, a scale-out OLTP/OLAP offering.

Now we will pivot to the key areas of Migration, Monitoring, High Availability, Backup, Security, Disaster Recovery, Compute, Memory and Storage & Licensing while comparing Azure SQL Database Offerings with Amazon RDS for SQL Server and RDS SQL Custom.

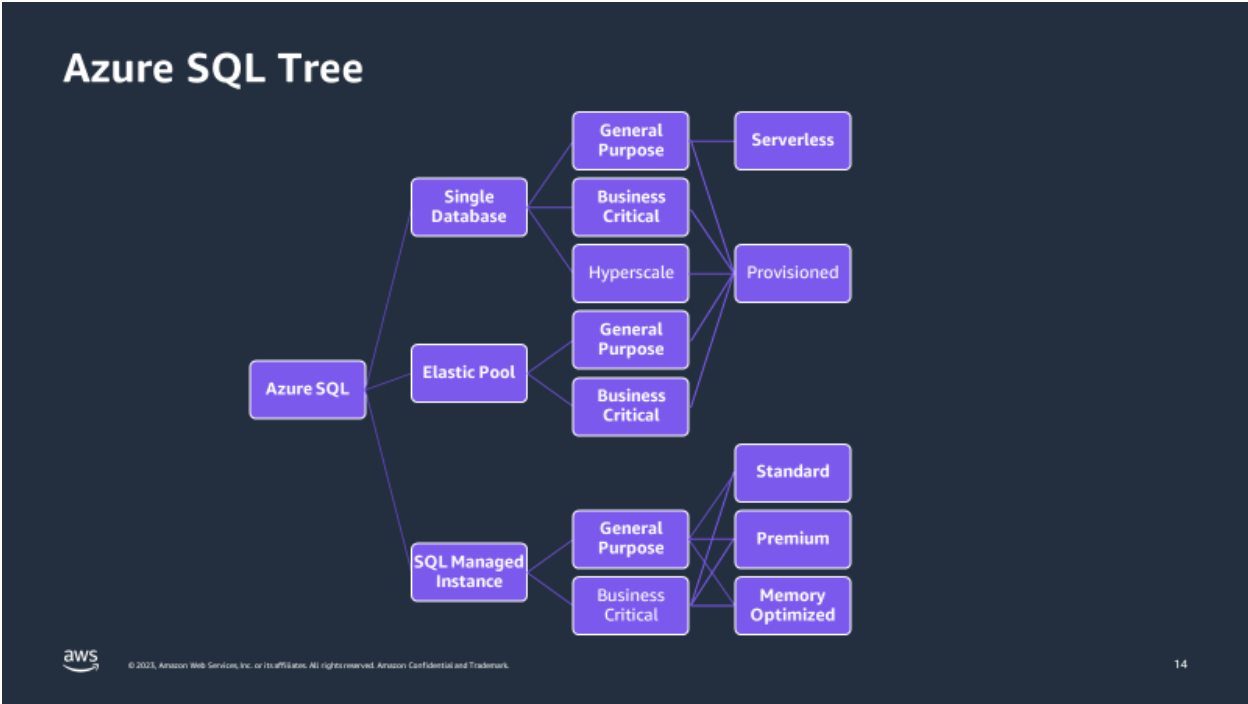
Deployment Models in Azure SQL

There are 6 different deployment options for SQL on Azure. On the far left, we have SQL Server on VMs, much like the EC2 and dedicated host options. On the far right, we have Azure SQL Edge, which is intended for scenarios like IoT, where the database is required to run on a small device outside the cloud. While we are here, I do want to reiterate that the focus of the document will be on the managed SQL options. If you compared SQL Server on Azure VMs against SQL on EC2, you would find a lot of similarity. The features are similar, the management is similar, even the compute cost is somewhat similar. Going forward, we will cover some of the differentiation in both directions, and that you will find in the managed offerings.



Azure SQL Tree

At the end of the day, this is what the tree might look like. For simplicity's sake, we have even removed Managed Instance Pools (which are in preview) and the DTU pricing model (which is not recommended). We have three different deployment options—a single database, a pool or resources shared by multiple, or a managed instance. In each case, we can choose from two or more service tiers, and then between serverless, provisioned, and other compute tiers. Please note that Serverless compute (despite being widely marketed) is only available for the General-Purpose tier of Microsoft Azure Single Database offering. In other words, even if a customer is excited about the potential of serverless, it may not meet their needs for performance or high availability.



Migration

Database migration is process of moving customer's database from a source instance to a destination instance. This process includes not only the move from one instance to another, but also a modernization process - from one version of SQL Server to a newer one (or from a legacy & unsupported to a supported one).

Azure Migration tools can be split into 2 main categories – *Online* and *Offline*. *Offline* migrations are those where execution of the migration involves downtime for the database access. For business-critical databases that can have no downtime, the situation can be mitigated by executing an *Online* migration.

The following table summarizes currently available methods to migrate to Azure SQL Managed Instance through offline methods and online methods

	A	B	C
1	Migration option	Features	Limitations/Notes
2	Backup and Restore	Supports all database sizes. Complete migration process only.	<i>Offline</i> migration.
3	ADF (Azure Data Factory)	Supports copying at table level. Partial or Complete migration processes.	<i>Offline</i> migration. Requires downtime that can be lowered by parallelizing the process. Database needs to be pre-created at destination Additional configuration to setup Self-hosted IR

4	Smart bulk copy	Supports copying data at database level. Partial or Complete migration processes.	<i>Offline</i> migration. Requires downtime that can be lowered by parallelizing the process. Database & schema need to be pre-created at the destination
5	Azure Data Migration Service	Database-sensitive migration service moves data, schema, and objects to Azure	Online Migration but Azure Database Migration Service doesn't support databases with multiple log files.
6	Managed Instance link feature	Enables near real-time data replication from SQL Server to Azure SQL Managed Instance.	Currently in Limited Preview . Managed Instance link feature uses distributed availability groups to extend your SQL Server on-premises Always On availability group hosted anywhere to Azure SQL Managed Instance in a safe and secure manner, replicating data in near real-time. And, with SQL Server 2022, the link feature enables disaster recovery.
7	Azure SQL migration extension for Azure Data Studio	Azure SQL migration extension for Azure Data Studio enables you to assess, get right-sized Azure recommendations and migrate your SQL Server databases to Azure.	Migration can be online (for migrations that require minimal downtime) or offline (for migrations where downtime persists throughout the migration) depending on your business requirements.
8	Log Replay Service	LRS is a free cloud service that's available for Azure SQL Managed Instance and based on SQL Server log-shipping technology.	When Azure Database Migration Service, or the Azure SQL extension for migration can't be used, LRS directly with PowerShell, Azure CLI cmdlets, or APIs can be used to manually build and orchestrate database migrations to SQL Managed Instance deployments.

Table 1: Online and Offline Migration tools in Azure

Now, let's see what available Migration tools/ feature available between Amazon RDS SQL Server & Azure SQL Services.

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL/Elastic Pool
2	Database Migration Service	Y	Y
3	Transaction replication	Y	Y

		(Subscriber Only)	
4	Native Backup/Restore	Y	N
5	SQL Server Log Shipping	Y (Custom log shipping)	N
6	CDC	Y	Y
7	Bacpac Method	Y	Y
8	Cloudbasix	Y	Y

Table 2: Migration Features/Tools comparison b/w RDS SQL Server & Azure SQL DB

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL Managed Instance
2	Database Migration Service	Y	Y
3	Transaction replication	Y (Subscriber Only)	Y
4	Native Backup	Y	Y (Copy Only)
5	Native Restore	Y	N
6	SQL Server Log Shipping	Y (Custom log shipping)	Y (Log Replay Service)
7	CDC	Y	Y
8	Cloudbasix	Y	Y

Table 3: Migration tools comparison between RDS SQL Server & Azure SQL Managed Instance

AWS Messaging

Microsoft Azure SQL offerings are complicated and customers are often confused about the various choices and service tiers they have to choose. When customers are migrating from Azure SQL to Amazon RDS for SQL Server, the mapping and translation between the two offerings become very difficult. Azure SQL have 2 cost based model 1) DTU (Database Throughput Unit & 2) vCORE Model (similar to vCPU in RDS). One thing to highlight about DTU is that it is a “License Included” model, which means Microsoft already factors in the SQL Server license cost into the price, and BYOL is not applicable to DTU. Another point is that, DTU is a On-Demand pricing model, so Reserved Capacity (equivalent to Reserved Instance on AWS) concept is not applicable here. Here is the basic rule of Thumb when migrating Azure SQL workload to RDS SQL Server.

- Every 100 DTUs in the Basic or Standard tiers can be mapped to 1 vCore.
- Every 125 DTUs in the Premium tier can be mapped to 1 vCore. **Note:** This rule is approximate because it does not consider the hardware generation used for the DTU database or elastic pool. In the DTU model, any available [hardware generation](#) can be used for your database or elastic pool.

For more details : <https://w.amazon.com/bin/view/RDS/SqlServer/Outbound/Blogs/how-to-right-size-dtu-based-sql-server>

AWS Partner product, Cloudbasix can also be considered for migrating data including TDE enabled databases from Azure SQL DB to Amazon RDS SQL Server. You can find a demo [here](#).

Azure Managed Instance have following limitations when working with backups and Azure SQL Managed Instance:

- Geo-restore of a database can only be performed to an instance in the same subscription as the source SQL managed instance.
- SQL Managed Instance databases can only be [restored to SQL Server 2022](#) (either on-premises, or on a virtual machine) if the source SQL Managed Instance has enrolled in the [November 2022 feature wave](#).
- SQL Managed Instance databases are encrypted with TDE by default. To restore your database to an instance other than the source SQL Managed Instance, the target instance must have access to the same key in Azure Key Vault used to encrypt the source database, or you must disable TDE encryption on the source database before taking the backup.

Azure Advantage

Azure has a clear advantage with new feature known as '[Managed Instance link feature](#)'. The Managed Instance link feature uses distributed availability groups to extend your SQL Server on-premises Always On availability group hosted anywhere to Azure SQL Managed Instance in a safe and secure manner, replicating data in near real-time. The link supports single node SQL Server instances without existing availability groups, and also multiple-node SQL Server instances with existing availability groups. Through the link, customer can use the latest benefits of Azure without migrating your entire SQL Server data estate to the cloud.

The link feature currently offers the following functionality:

One-way replication (SQL Server versions 2016 and 2019): Use the link feature to replicate data one way from a SQL Server instance to your managed instance. Although you can manually fail over to your managed instance if there's a disaster, doing so breaks the link, and failing back isn't supported.

Disaster recovery (SQL Server 2022): Use the link feature to replicate data from a SQL Server 2022 instance to your managed instance, manually fail over to your managed instance during a disaster, and fail back to the SQL Server instance after you've mitigated the disaster. Though this feature is currently in **limited preview** but this will significantly help Azure customer to fast track their migration journey to Azure SQL Managed instance and also help maintaining a DR site in Azure keeping primary SQL Server on-premises.

Another benefit customer has during migration is Log Replay Service. [LRS](#) is a free cloud service that's available for Azure SQL Managed Instance and based on SQL Server log-shipping technology. LRS further enables complex custom migrations and hybrid architectures between on-premises SQL Server instances and SQL Managed Instance deployments.

When you can't use Azure Database Migration Service, or the Azure SQL extension for migration, you can use LRS directly with PowerShell, Azure CLI cmdlets, or APIs to manually build and orchestrate database migrations to SQL Managed Instance deployments.

Reference Links

[Migrate SQL Server databases from an Azure SQL database to Amazon RDS for SQL Server using bacpac method | AWS Database Blog](#)

[Using a Microsoft SQL Server database as a source for AWS DMS - AWS Database Migration Service \(amazon.com\)](#)

[Using a Microsoft SQL Server database as a target for AWS Database Migration Service - AWS Database Migration Service \(amazon.com\)](#)

[Migrating your SQL Server database to Amazon RDS for SQL Server using AWS DMS | AWS Database Blog](#)

[Migrating SQL Server databases from Microsoft Azure to AWS in near-real time with CloudBasic | AWS Database Blog \(amazon.com\)](#)

[Migrate databases by using Log Replay Service - Azure SQL Managed Instance | Microsoft Learn](#)
<https://learn.microsoft.com/en-us/azure/azure-sql/managed-instance/restore-sample-database-quickstart?view=azuresql>

[Restore a database from a backup - Azure SQL Managed Instance | Microsoft Learn](#)

Monitoring

To monitor the performance of a database in Azure SQL Database and Azure SQL Managed Instance, multiple native tools are available to provide a comprehensive view. Microsoft also claims that Azure SQL Database and Azure SQL Managed Instance provide advanced monitoring and tuning capabilities backed by artificial intelligence to assist in troubleshooting and maximizing the performance of databases and solutions.

	A	B	C
1	Monitoring Solution	Azure SQL Database	SQL Managed Instance
2	Monitor using DMVs	Yes	Yes
3	Monitor using query store	Yes	Yes
4	SQL Insights (preview) in Azure Monitor	Yes	Yes
5	Azure SQL Analytics (preview) using Azure Monitor Logs *	Yes	Yes

Table 4: Monitoring Tools available in Azure

Database monitoring is one of the important tasks of a database administrator to ensure availability and performance. AWS also provides different tools to monitor the database instance. Below are some of the options.

- Amazon RDS Events
- Enhanced Monitoring
- CloudWatch metrics
- CloudWatch Alarms
- Database log files

Now, let's see what available Monitoring tools/ feature available between Amazon RDS SQL Server & Azure SQL Services.

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL /Elastic Pool / Azure Managed Instance
2	Fine Grained Performance monitoring, with dashboard and long-term retention	Y	Y
3	Monitoring DB Instance OS*	Y	Y
4	Ability to setup notification on database related events	Y	Y
5	Database Alerts	Y	Y
6	Auto Tuning SQL	N	Y
7	Individual Database Cost	N	Y
8	DMV Support	Y	Y

Table 5: Monitoring Options comparison b/w RDS SQL Server & Azure SQL Services (all)

Note *: AWS Performance Insights allows to monitor certain specific OS metrics .For more details please refer to link [here](#).

Reference Links:

[Best Practices on how to configure Monitoring and Alerts for Amazon RDS for SQL Server: Part1 | AWS Database Blog](#)

[Monitoring metrics in an Amazon RDS instance - Amazon Relational Database Service](#)
<https://learn.microsoft.com/en-us/azure/azure-sql/database/automatic-tuning-enable?view=azuresql>

Azure Advantage

Azure provides deep monitoring features and provides further integration with SQL insights which is currently in Preview. SQL Insights (preview) is a comprehensive solution for monitoring any product in the [Azure SQL family](#). SQL Insights uses [dynamic management views](#) to expose the data that you need to monitor health, diagnose problems and tune performance. You can find more details about SQL insights [here](#).

AWS Messaging

Overall RDS has multiple Cloud native tools to provide a comprehensive monitoring of SQL Server. However, we do not provide cost breakdown of individual databases and Auto Tuning capabilities of SQL Server Databases similar to Azure SQL. Having said that, automatic tuning feature of Azure SQL Database is pretty basic and recommendations are not too aligned during complex TSQL query optimization. In RDS SQL Server rebuild index is generally not required due to the way EBS storage is designed. You can still do rebuild index and it can be automated through SQL Agent jobs. Regarding missing index or unused index recommendation, this still can be worked out by providing custom T SQL scripts which can be scheduled once through SQL Agent job. Going back to other tuning opportunities, with SQL Server 2017 and beyond on RDS, it by default provides self-tuning capabilities in the database engine itself. And this is what even Auto tuning feature in Azure uses.

Security

Overall, both AWS & Azure provides security strategies to protect database with layered defense-in-depth approach providing networking security, access management and encryption service for data protection. Therefore, have very similar concepts of VPC, Security groups, Network ACL to protect the database environment.

But Azure has made progress to help customers improve their overall security strategies and posture by providing security related features natively on Azure Platform. These capabilities address the needs of the customers for secure database access, implement threat detection and vulnerability scanning solution.

Azure AD now supports windows authentication for SQL Server

It is important to highlight that Azure has now enabled customers to use Azure AD for SQL Server authentication. To use Microsoft 365, cloud-based collaboration and productivity tools, customers required Azure Active Directory (Azure AD). Azure AD was a cloud-based directory to manage user identities allowing them to integrate their on-premises Active Directory and set up single sign-on feature to access office business application and other apps.

Apart from Office application, customers also used Azure AD for multi-factor authentication and few more security features such conditional access policy to restrict users from accessing from specific site, ip addresses block etc. Despite of Azure AD, customers still had to deploy [Azure ADDS](#) (Azure active directory domain services) to support windows authentication for SQL Server. Azure AD now extended that capabilities to support kerberos authentication for Windows and SQL Server authentication. Customers now don't need to manage Active directory on a Windows Server as they did for years on-premises to support kerberos authentication. Azure position this feature as an effort to modernize the security infrastructure. Refer to [Windows Authentication for Azure AD Principals or SQL Managed Instance](#).

From database administration standpoint, this adds one extra layer of additional security when connecting to database instance using SSMS. DBAs now will be challenged for [multi-factor authentication](#) capabilities in addition to Windows authentication when connecting using SQL Server Management Studio. In addition to this, security administrators can use [conditional access policy](#) to implement more granular policy to secure the infrastructure and user access to the database.

Another observation about Azure SQL offerings is that TDE (Transparent Data Encryption) is enabled by default. Enabled by default for Azure SQL DB. Applied the instance therefore all the databases inherit at the startup. Customer also have the ability to bring customers own key (BYOK) and manage their keys using azure key vault services. Azure key vault services provide geo-redundancy capabilities as well.

AWS Advantage

None.

Reference Links:

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL Managed Instance
2	Encryption at-rest	Y	Y
3	Encryption at-rest with customer managed keys	Y	Y
4	Encryption in-transit	Y	Y
5	Supports SQL Server Row Level Security (RLS) policies	Y	Y
6	TDE	Y	Y
7	Always Encrypted	Y	Y
8	Always Encrypted with Secure Enclaves	N	Y
9	Secret Manager	Y	Y
10	Dynamic Data Masking (DDM)	Y	Y
11	Multi-factor Authentication	N	Y
12	Threat protection & Vulnerability assessment	N	Y
13	SQL Server Audit	Y	Y

Table 6: Security Features between RDS SQL Server & Azure SQL Managed Instance

[Reference Links]

[SQL Advanced Threat Protection](#)

[Auditing for Azure SQL Database](#)

[Overview of Azure SQL Managed Instance security](#)

[Advanced Threat Protection in Azure SQL Managed Instance](#)

[Configure multi-factor authentication for SSMS and Azure AD](#)

[Azure AD authentication for SQL Server](#)

[Multi-factor Authentication](#)

[Enable Transparent Data Encryption with customer-managed key from Azure Key Vault](#)

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL and Elastic Pool
2	Encryption at-rest	Y	Y
3	Encryption at-rest with customer managed keys	Y	Y
4	Encryption in-transit	Y	Y
5	Supports SQL Server Row Level Security (RLS) policies	Y	Y
6	TDE	Y	Y
7	Always Encrypted	Y	Y
8	Always Encrypted with Secure Enclaves	N	Y
9	Secret Manager	Y	Y
10	Multi-factor Authentication	N	Y
11	Threat protection & Vulnerability assessment	N	Y
11	SQL Server Audit	Y	Y

Table 7: Security Features between RDS SQL Server & Azure SQL/Elastic Pool

AWS Messaging:

Amazon RDS for SQL Server provides all enterprise grade Security Feature for hosting sensitive workloads. Currently AWS customers can use 3rd Party Services which are available in AWS Marketplace like [IBM Guardium](#) for threat detection & Vulnerability Assessment .Also, Amazon Relational Database Service (Amazon RDS) for SQL Server now supports database activity streams (DAS) to provide a near-real-time stream of database activities in your relational database. To protect database from internal and external threats, and to meet compliance and regulatory requirements, you can easily integrate the database activity stream with third-party database activity monitoring tools to monitor and audit database activities. Our overall story with KMS, TDE and Encryption in Transit provides end to end security to host customer data across industry verticals.

Compute, Memory & Storage

Azure provides 2 purchase models: vCore and DTU Based. Since Amazon RDS MS SQL Server is most similar to the vCore model, the below comparison is based on Azure SQL with vCore vs Amazon RDS MS SQL Server with instance class types.

	A	B	C
1	Managed Relational Database: compatible with SQL Server	<u>Amazon RDS for SQL Server</u>	<u>Azure SQL Managed Instance /Next Gen MI</u>
2	BYOL (Bring your own License)	No	Yes

3	Deployment	Single AZ, Multi-AZ	General Purpose/Business Critical tier/Next Gen
4	Versions supported	All supported versions till SQL 2022	All supported versions till SQL 2019
5	Editions supported	STD, ENT, Web, Express	STD, ENT
6	Database size (max)	16 TB (64 TB in private preview)	16 TB/ 32 TB (Next-Gen)
7	RAM per database instance (max)	3904 GB	870 GB
8	Custom Instance Size	N	Y
9	vCPUs per database instance (max)	1-128 vCPUs	Upto 80 vCores/ 128
10	Provisioned IOPS: ability to configure IOPS	Y	N
11	DB Concurrent Connections	30000	30000
12	Number of Databases per Instance	100	100/500
13	Scaling: horizontal (scale-out SQL Enterprise Edition)	Y	Y
14	Max IOPS(for Next Gen GP)	UPTO 64000 IOPS	Per instance: 192 MB/s Per DB: no limits! (Next Gen)/ 80k IOPS
15	Scaling: storage scaling capability	Y	Y
16	Scaling: storage scaling capability (independent and automatic)	Y	Y
17	Scaling: vertical (scale-up)	Y	Y
18	Kerberos Authentication	Y	Y
19	Native SQL Reporting Service (BI)	Y	N
20	Native SQL Integration Service (BI)	Y	N
21	Native SQL Analysis Services (BI)	Y	N
22	SLA	99.95	99.99

Table 8: Storage, Compute & Memory comparison b/w RDS SQL & Azure Managed Instance

	A	B	C
1	Managed Relational Database: compatible with SQL Server	<u>Amazon RDS for SQL Server</u>	<u>Azure SQL Single DB</u>
2	Max number of databases	100	1
3	MAX storage/per Database	16 TB (64 TB in Preview)	general purpose: or Business critical service tiers gives max: 4 TB Hyperscale gives max: 100 TB

4	Memory and vCPUs	Broad range of Instance class types which provides a non-variable hardware configuration with different class types as standard, memory optimized and burstable with number of vCPUs up to 128 and max memory up to 3904 GB with max 16 TB	Based on service tiers (general purpose, business critical and hyperscale) which provides a variable hardware configuration with number of vCore up to 128 and Memory up to 625 GB
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Table 9: Storage, Compute & Memory options comparison b/w RDS SQL & Azure SQL (Single DB).

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure Elastic Pool
	Max number of databases	100	Up to 5000
2	MAX storage	16 TB (Private Preview 64 TB)	Max size varies from 1- 4 TB based on number of vCores chosen
3	Memory and vCPUs	Broad range of Instance class types which provides a non-variable hardware configuration with different class types as standard, memory optimized and burstable with number of vCPUs up to 128 and max memory up to 3904 GB with max 16 TB.	a variable hardware configuration with number of vCore up to 128 and Memory up to 625 GB.

Table 10: Storage, Compute & Memory options b/w RDS SQL & Azure SQL Elastic Pool

AWS Advantage

AWS offers several types of [instance classes](#) for I/O, Memory and general-purpose workloads with max size up to 16 TB (along with private preview program of 64 TB) with 100 databases per instance. Recently we have announced a [Private Preview Program](#) where we are now

supporting hosting 200 tenants/databases in SAZ environment and also for Production workloads. RDS SQL Server provides higher CPU to memory ratio with X instance class compare to Azure SQL. Also, AWS customers can use Provisioned IOPS on RDS SQL where in Azure customer have to scale higher tier to get more IOPS & throughput which eventually increase the cost. As numbers are always subject to change, for more latest update please refer to resource limit of [Azure SQL DB](#) and [Azure SQL Managed Instance](#).

Also it is important to highlight to our customers that RDS SQL Server supports SQL native BI components (SSIS, SSAS, SSRS) whereas Azure SQL Managed Instance doesn't. If customer wants to use SQL BI comments on Azure they are separate BI Services and they will incur additional costs.

Azure Advantage

Azure has significant advantage here as it gives customer flexibility to host a single database and pay for just one database rather than for whole SQL instance. With Azure SQL Elastic Pool, customer have significant advantage to host upto 5000 databases (mostly for multitenant SAAS applications) in a single pool in a cost effective & secured manner . To host similar number of databases we can recommend customer to use RDS SQL Custom however our observations are RDS SQL Custom with License Included Model is expensive unless customer have their own existing license with Software Assurance (SA). will RDS SQL Custom with BYOM and reserved Instance Pricing can be a cost-effective solution however customer lose the ability to scale compute to a different RDS SQL instance type due to the restrictions of using Reserved Instance on RDS SQL Server for atleast 1 year. In summary, Microsoft SQL Elastic pool, which is a complete sharding framework that allows customer to manage shard maps, create shards, execute query fan-out, data-dependent-routing and shard rebalancing. These features gives Azure SQL Database a distinct advantage in the eyes of many SaaS providers. Recently , the announcement of Private Preview Program of supporting 200 tenants/databases in a single RDS SQL Server instance in a SAZ environment will bridge gap eventually however we need data to confirm that . We strongly recommend to have a true alternative of Azure SQL Elastic Pool for our customers and eventually helps us to grow marketshare for RDS SQL Server.

AWS Messaging

RDS SQL Custom with BYOM and reserved Instance Pricing can be a cost-effective solution. If customer have existing SQL License with Software Assurance then RDS SQL Custom should be our first approach in a compete scenario.

High Availability

High availability offerings are different between AWS and Azure. Azure has more advantages compared to RDS.

Azure Advantage:

Azure provides Local and Zone redundant high availability for its databases. The local redundant availability model's architecture for basic, standard, and general-purpose service tier is based

on the separation of the compute and storage layers to ensure data availability; it is similar to failover cluster instances (FCIs).

The Zone redundant availability model for general-purpose service tier uses Azure Availability Zones to replicate a database across multiple physical locations within the same Azure region. Utilizing zone-redundant storage ensure that the data will be copied synchronously three times across three Availability Zones in the same region. However, the zone-redundant configuration is currently not available in SQL Managed Instance.

In the premium/business-critical tier locally redundant availability model, High availability is achieved by replicating both compute and storage to additional nodes creating a three to four-node cluster. It is implemented by technology similar to AOAG.

The premium/business critical tier Zone redundant configuration is similar to local redundant availability model except that the replicas are placed across different Availability Zones within the same region. This feature is currently in preview for Azure SQL Managed Instance, and only available on the business-critical service tier. In SQL Database, when using the business-critical tier, zone-redundant configuration is only available when the Gen5 hardware is selected.

Azure SQL guarantees up to [99.99%](#) availability of service. It means service can be down only for 4 minutes in a month or 52 minutes in a year v/s RDS which can be down up to 21 minutes a month or 4 Hour 20 minutes in a year.

The premium/business-critical tier locally redundant availability model for Azure SQL database provides access to one internal secondary replica to offload the read workload using AOAG technology.

AWS Advantage:

Automatic Failover duration is less (~few minutes) with RDS for SQL Server when using Multi-AZ. In Azure, customers can recover from an outage if they have configured either geo-secondary using [active geo-replication](#) or an [auto-failover group](#) for customers database or databases. Depending on customers choice of these technologies, customers can use either manual or automatic failover. While failover itself takes only a few seconds, the service will take at least 1 hour to activate it. This is necessary to ensure that the failover is justified by the scale of the outage. Also, the failover may result in small data loss due to the nature of asynchronous replication.

[Reference Links]

[Azure High Availability](#)

[Geo-Replication v/s Failover Group Comparison](#)

[Azure RTO and RPO](#)

The following tables compares different high availability options between Amazon RDS for SQL Server and Azure SQL as well as Azure Managed Instances.

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL/Elastic Pool

2	High Availability (Master/Slave) configuration	Y	Y
3	Enable and disable high availability (Existing instance)	Y	N
4	SLA	99.95	99.99
5	Failover (automated) capability In-Region	Y (Multi-AZ)	Y
6	Failover test (manual) capability	Y	Y
7	Automatic failover duration	Few minutes	1 Hour
8	RTO	Depends on redo log volume (Usually minutes)	30 sec (SQL DB Business Critical tier with Active Geo-Replication)
9	RPO	~0 (With Multi-AZ)	5 sec (SQL DB Business Critical tier with Active Geo-Replication)

Table 11: HA Options b/w RDS SQL & Azure SQL Elastic Pool

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL Managed Instance
2	High Availability (Master/Slave) configuration	Y	Y
3	Enable and disable high availability (Existing instance)	Y	N
4	SLA	99.95	99.99
5	Failover (automated) capability In-Region	Y (Multi-AZ)	N
6	Failover test (manual) capability	Y	Y
7	Automatic failover duration	Few minutes	1 Hour
8	Ability to managed database connection using proxy	Y	Y
9	RTO	Depends on redo log volume (Usually minutes)	NA
10	RPO	~0 (With Multi-AZ)	NA

Table 12: HA Options b/w RDS SQL & Azure SQL Managed Instance

Cross-region Disaster Recovery

Customer require both high availability and disaster recovery to protect their business-critical applications. They are looking for bi-directional replication solution, easy to maintain copies of backups in more than one region to support their disaster recovery requirement.

For business continuity and disaster recovery, Azure provides capabilities listed below for both Azure SQL DB and Azure Managed SQL Instance.

1. Automated backup
2. Active-Geo Replication
3. Auto-failover Groups

Azure Advantages

For [automated backup](#), Azure provides [backup storage redundancy](#) for both Azure SQL DB and SQL Managed Instance. Customers simply choose the one the options below. **Geo redundant storage** (GRS) provides cost-effective DR capabilities for customers.

For database replication, Azure SQL DB customers can use active-geo replication [active geo-replication](#) feature to create readable secondary database both in-region and cross-region using async replication. Applications can easily recover from the complete or partial loss of an Azure region due to a natural disaster, catastrophic human errors. Customers can also use active-geo replication for other use-cases such as to support database migration between Azure regions and [managing rolling application upgrades](#). Azure recommends customers to use active-geo replication along auto-failover group to make their DR failover and failback seamless. Active geo-replication isn't part of Azure SQL MI however customers can use [auto-failover groups](#) to achieve the same result. Azure SQL Managed Instance supports auto-failover groups for geo-replicated secondary SQL Managed Instance with user-configured failover policy.

Azure provides capability where the read-write and read-only listener **end-points remain unchanged** during geo-failovers using **Auto-Failover group feature**. After the geo-failover is completed, the DNS record is automatically updated to redirect the endpoints to the new region.

Also, Azure SQL Managed Instances supports being publisher using [MS Transactional replication](#). Supports both publisher with local and remote distributor on SQL Managed Instance.

AWS Advantages

None.

References

[Manage rolling upgrades of cloud applications by using SQL Database active geo-replication](#)

[Designing globally available services using Azure SQL Database](#)

[Disaster recovery strategies for applications using Azure SQL Database elastic pools](#)

[Auto-failover groups overview & best practices \(Azure SQL Database\)](#)

[Configure and failover a single database using active geo-replication](#)

[Configure and failover a pooled database using active geo-replication](#)

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL Managed Instance

2	Read Replicas (in-region)	Y	Y
3	Read Replicas (cross-region) (SQL Server Enterprise- 2017 & 2019)	Y	Y
4	Number of read replicas supported	5 (Enterprise edition)	0 replica (General purpose) 1 replica (Business critical tier)
5	Automatic Failover	N	Y
6	Failover capability Cross-Region	Y (Promote Read Replica)	Y (Using Failover Groups)
7	Endpoints after Cross Region failover	Different	Same (Only with Auto-Failover Groups) Active Geo-Replication NOT Supported
8	Failback to Original Region	Manual	Automated
9	Supports Sub/Pub model for database migrations (transactional replication)	Only Subscriber	Publisher & Subscriber

Table 13: DR feature b/w RDS SQL & Azure SQL Managed Instance

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	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL & Elastic Pool
2	Read Replicas (in-region)	Y	Y
3	Read Replicas (cross-region)	Y	Y
4	Number of read replicas supported	5	4
5	Automatic Failover	N	Y
6	Failover capability Cross-Region	Y (Promote Read Replica)	Y (Using Failover Groups)
7	Endpoints after Cross Region failover	Different	Same (Only with Auto- Failover Groups) Different if Active Geo-Replication
8	Failback to Original Region	Manual	Automated
9	Supports Sub/Pub model for database migrations (MS transactional replication)	Only Subscriber	Only Subscriber

Table 14: DR feature b/w RDS SQL & Azure SQL /Elastic Pool

Backup & Recovery

Database backups helps customers recover from system outages, unwanted deletions or updates, database corruption issues, and other related issues.

AWS Advantage

AWS provides instance level snapshots for restoring a SQL Server database instance as well as for PITR.

Customers can perform a native backup using RDS stored procedures. In Azure SQL Database, conventional database backup statements don't work. A manual backup consists of exporting the database as a DACPAC (schema) or BACPAC (schema + data) and BCP (bulk copy program utility) out the data into CSV files.

User initiated SQL Server copy only backups from an Azure SQL Managed Instance can only be restored to other Azure SQL managed instances, or to SQL Server 2022. They cannot be restored to other versions of SQL Server, or to Azure SQL Database.

Customers can restore an RDS database backup to another server/instance. An Azure SQL Database can only be restored on the same Azure SQL Server as the original database with a

different name. On Azure SQL Managed Instance, customers can perform a restore on the same database or a restore on a different instance in the same region and subscription.

Azure Advantage:

Azure SQL Managed Instance provides [PITR recovery](#) for individual User Databases. By default, Azure SQL Database stores backups in geo-redundant [storage blobs](#) that are replicated to a [paired region](#). Geo-redundancy allows customers to restore their databases in a different region in the event of a regional outage.

Customers can choose one of the following storage redundancies for backups:

Locally redundant storage (LRS): Copies customers backups synchronously three times within a single physical location in the primary region. LRS is the least expensive storage option, but we don't recommend it for applications that require resiliency to regional outages or a guarantee of high data durability.

Zone-redundant storage (ZRS): Copies customers backups synchronously across three Azure availability zones in the primary region. It's currently available in only [certain regions](#).

Geo-redundant storage (GRS): Copies customers backups synchronously three times within a single physical location in the primary region by using LRS. Then it copies customers data asynchronously three times to a single physical location in the [paired](#) secondary region. It is **the most expensive option** among the three.

In Azure, the backups are TDE encrypted by default.

Also, customers can perform **point-in-time restores (PiTR)** of **individual databases** on Azure SQL Database and Azure SQL Managed Instance. Customers can use the Azure portal, PowerShell, the Azure CLI, or the Azure SDK. PiTR uses the automatic Full, Differential, and Log backups.

Reference Links

[Azure backup storage redundancy](#)

[Azure restore features](#)

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure SQL/Elastic Pool
2	Ability to take Snapshots	Y	N
3	Snapshots/Backups: ability to copy snapshots/backups across regions	Y	Y(Backups)
4	Snapshots/Backups: ability to encrypt snapshots/backups	Y	Y(Backups)
5	Snapshots/Backups: automatic capability, with Point-In-Time Recovery (PITR)	Y	Y
6	Snapshots/Backups: on-demand/manual/user-initiated capability	Y	N

7	SQL Server Backup	Y	Y
8	Native SQL Server Backup	Y	N
9	Backup Storage Redundancy	Default Zone Redundant	Default Geo-redundant
10	Ability to set retention (Short & long term)	Y	Y
11	Customize the location for automated & manual snapshot backup	N	N
12	Frequency: Automated snapshot/backup	Daily Snapshot + Tlog at 5 mins Interval	Weekly Fulls, 12 or 24 Hrs Diffs and TLog every 5-10 mins

Table 15: Backup & Recovery Features b/w RDS SQL Server & Azure Single DB

	A	B	C
1	Managed Relational Database: compatible with SQL Server	Amazon RDS for SQL Server	Azure Managed Instance
2	Ability to take Snapshots	Y	N
3	Snapshots/Backups: ability to copy snapshots/backups across regions	Y	Y(Backups)
4	Snapshots/Backups: ability to encrypt snapshots/backups	Y	Y(Backups)
5	Snapshots/Backups: automatic capability, with Point-In-Time Recovery (PITR)	Y	Y
6	Snapshots/Backups: on-demand/manual/user-initiated capability	Y	N
7	SQL Server Backup	Y	Y
8	Native SQL Server Backup	Y	N
9	Backup Storage Redundancy	Default Zone Redundant	Default Geo-redundant
10	Ability to set retention (Short & long term)	Y	Y
11	Customize the location for automated & manual snapshot backup	N	N
12	Individual Database Pointintime Restore (PITR)	N	Y
12	Frequency: Automated snapshot/backup	Daily Snapshot + Tlog at 5 mins Interval	Weekly Fulls, 12 or 24 Hrs Diffs and TLog every 5-10 mins

Table 16: Backup & Recovery Features b/w RDS SQL Server & Azure SQL Managed Instance

Licensing

Lastly, licensing is a frequent tactic Microsoft uses to compete with RDS. License Mobility is not supported in RDS, and Microsoft has an Azure Hybrid benefit that allows customers to bring on-premise licenses to the cloud. Remind the customers that Azure Hybrid benefit requires Software Assurance (which is a significant extra cost they must pay every year, perpetually). It is also not the overall TCO of a SQL server, so try the tips in the recommendations section below to compete on price.

For customers considering a migration, run an Optimization and Licensing Assessment (OLA). The OLA can help the customer and the account team understand usage and optimize compute and licensing costs. For customers considering a cloud migration, also be sure to consider available programs, such as the [Migration Acceleration Program \(MAP\)](#). Customers can receive funding and credits to accelerate their move and offset upfront costs. MAP even provides additional benefits for customers migrating [specific workloads](#), including Microsoft SQL Server. If the customer is interested in modernizing their databases, run the [Schema Conversion Tool](#) to see if their current implementation could be migrated to PostgreSQL or MySQL. Additionally, they could run the [Compass script](#) to test compatibility with Babelfish for PostgreSQL. Migrating to a managed service running an Open Source database engine would help modernize the customer's data tier while freeing them from restrictive and expensive SQL Server licenses.

Reference: [Cloud Intelligence - Microsoft Azure SQL \(amazon.com\)](#)

Azure Advantage

[Azure Hybrid Benefit](#) allows customers to exchange their existing licenses for discounted rates on Azure SQL Database and Azure SQL Managed Instance. Customers can save up to 30 percent or more on SQL Database and SQL Managed Instance by using your Software Assurance-enabled SQL Server licenses on Azure. The [Azure Hybrid Benefit](#) page has a calculator to help determine savings. With Azure Hybrid Benefit, they pay only for the underlying Azure infrastructure by using your existing SQL Server license for the SQL Server database engine itself (Base Compute pricing). If they do not use Azure Hybrid Benefit, they pay for both the underlying infrastructure and the SQL Server license (License-Included pricing).

License footprint	What does Azure Hybrid Benefit for SQL Server get you?
SQL Server Enterprise Edition core customers with SA	• Can pay base rate on Hyperscale, General Purpose, or Business Critical SKU • One core on-premises = Four vCores in Hyperscale SKU • One core on-premises = Four vCores in General Purpose SKU • One core on-premises = One vCore in Business Critical SKU
SQL Server Standard Edition core customers with SA	• Can pay base rate on Hyperscale, General Purpose, or Business Critical SKU • One core on-premises = One vCore in Hyperscale SKU • One core on-premises = One vCore in General Purpose SKU • Four cores on-premises = One vCore in Business Critical SKU

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680 [Azure SQL Database Free Tier](#) gives customers a fully featured Azure SQL serverless database
681 with a free amount of 100,000 vCore seconds of compute and 32 GB of storage plus 32 GB of
682 backup storage every month, for the lifetime of your subscription. Customers can host one
683 database per Azure subscription and it offers access to the same features and capabilities as the
684 General Purpose tier. They can choose from .5 vCore to 4 vCore with the free amount and scale
685 up or down as needed. Customers can switch to continue usage mode and keep using their
686 database for additional charges allowing them to use their existing data, schema, settings, and
687 connections and seamlessly transition from free to paid usage.

688 **Limitations of Azure SQL Database Free Tier**

689 This feature is rolled out per region and may limit the availability and performance of
690 customers database depending on their location. The offer is limited to one database per Azure
691 subscription. If customers need more than one database, they will have to pay for additional
692 databases or create multiple subscriptions. It does not include some features that are available

in other service tiers, such as Hyperscale, Business Critical, or Premium. For example, you cannot use columnstore indexes, in-memory OLTP, or zone redundancy with the free offer.

With Azure SQL Managed Instance free tier customer can try Azure SQL MI free of charge for the first 12 months. They get a General Purpose instance with up to 100 databases, 720 vCore hours of compute every month and 64 GB of storage (including system files).

Limitations of Azure SQL Managed Instance Free Tier

Only 1 instance per subscription. Compute can be 4 and 8 vCore instances only. As with Azure SQL Database free tier, it is available in selected regions. IT is not currently possible to upgrade the free instance to the paid version and there is no guaranteed SLA>

Reference Links: [Azure Hybrid Benefit - Azure SQL Database & SQL Managed Instance | Microsoft Learn](#)

[Save compute costs with reserved capacity - Azure SQL Database & SQL Managed Instance | Microsoft Learn](#)

[Try Azure SQL Managed Instance for free \(preview\)](#)

Azure Hyperscale

Serverless for Hyperscale in Azure SQL Database brings together the benefits of serverless and Hyperscale into a single database solution. Now both compute and storage automatically scale based on workload demand for databases requiring up to 80 vCores and 100 TB. Billing is based on the amount of resources used rather than the amount provisioned. Serverless Hyperscale optimizes price-performance and simplifies performance management for databases with intermittent and unpredictable usage while leaving room for growing to extreme scale and delivering high performance during active usage periods.

Serverless Hyperscale automatically scales compute and cache resources in response to changing workload demand of the database. The primary replica and any high availability (HA) replicas or named replicas each auto-scale independently for a serverless database in Hyperscale. This auto-scaling independence adapts to workload variability that can occur across replicas based on their application purpose and optimizes resource allocation accordingly across the entire Hyperscale footprint. The compute billed for each replica is based on the amount of CPU and memory used per second between a configurable compute range.

Serverless Hyperscale contrasts with provisioned compute Hyperscale which allocates a fixed amount of compute resources for a fixed price and is billed per hour. Over short time scales, provisioned compute databases must either over-provision resources at a cost in order to accommodate peak usage or under-provision and risk poor performance. Over longer time scales, provisioned compute databases can be rescaled, but this solution may require predicting usage patterns or writing custom logic to trigger rescaling operations based on a schedule or

performance metrics which adds development and operational complexity. In serverless, compute scaling within configurable limits is managed natively by the service to continuously and quickly right-size resources.

Reference : [General availability: serverless for Hyperscale in Azure SQL Database - Microsoft Community Hub](#)

AWS Messaging:

We haven't seen any workload compete oppty b/w Azure Hyperscale and RDS SQL yet. Very recently Microsoft made SQL License free on Azure hyperscale to compete with Aurora PostgreSQL. We don't have any specific messaging however please open a Spereq to Database specialist team or reach out to Sudarshan Roy (roysudar@amazon.com) / Sudhir Amin (Sudhamin@amazon.com) and Minesh Chande (mineshch@amazon.com) for any compete oppty.

Customer Success Stories & Observations:(Compete with Azure)

Prime Focus Technologies (PFT) is the creator of Enterprise Resource Planning (ERP) software, CLEAR, for the Media & Entertainment (M&E) industry. It offers streaming platforms, studios, and broadcasters transformational AI-led technology and media services powered by the cloud that help them lower their Total Cost of Operations (TCOP) by automating their content supply chain. PFT works with M&E companies such as Walt Disney-owned Star TV, Channel 4, ITV, Sinclair Broadcast Group, A&E Networks, Warner Media, PBS, Hearst Television, CBS Television Studios, 20th Century Fox Television Studios, Lionsgate, Showtime, HBO, NBCU, TERN International, Disney+ Hotstar, BCCI, Indian Premier League and more.

With AWS, PFT can easily scale its storage within minutes, instead of weeks, to onboard new customers or respond to additional storage requests from existing customers. To process and store media content at scale, PFT deployed [Amazon FSx for Windows](#) and [Amazon FSx for Lustre](#) with [Amazon Simple Storage Service](#) (Amazon S3). [Amazon FSx](#) meets the performance needs for PFT's persistent and work-in-progress storage, and ensures the content is protected and highly available with data replication across AWS Regions.

The company has also improved its disaster recovery capabilities while saving about 30 percent on long-term storage. PFT uses a mix of [Amazon S3 Glacier storage classes](#) for long-term, durable, cost-effective data archival and fast retrieval of content. Within a few months of the migration, PFT digitized and uploaded 8-9 petabytes of data from tape storage to the cloud. PFT has also freed 10 percent of time previously spent on routine maintenance. The company runs [Amazon Relational Database Service for SQL Server](#) to automate administrative tasks such as backups, point-in-time snapshots, and security policies. Furthermore, PFT upgraded its now-deprecated SQL Server 2008 R2 database to a new version with minimal compatibility issues and changes.

Reference: [Prime Focus Technologies Case Study | Amazon Web Services](#)

FIS (Fidelity Investment Services)

Fidelity National Information Services is a US-based financial services technology provider. FIS Global has access to AWS's support for their business digital transformation & innovation needs. They had multi SQL Instance onprem hosting critical production applications which they want to move to either Azure or AWS Cloud. After going through multiple deep dive session on RDS SQL Server & Migration planning & Assessment, customer decided to move their workload to RDS SQL Server due to overall value proposition of RDS and integration with other Cloud Native Services

Infionic India: We engaged with Infionic team to move from Azure SQL DB Elastic Pool to RDS SQL Server. We strategically inserted our partner One Cloud Hub to position the ultimate aim of moving to PostgreSQL ensuring the database freedom story. During initial Migration Planning we did a comprehensive POC on RDS SQL Server and demonstrate faster query performance and ease of database platform management. Customer initial was apprehensive moving out of Azure SQL Elastic Pool due to auto tuning capabilities and ease of management. However, during POC we were able to mitigate those challenges. This document also covers those messaging element which helped us to shortlist RDS SQL Server as preferred Database Platform. Customer also agreed to be publicly referenceable.

Zee TV India:

ZEE Entertainment is among the largest producers and aggregators of Hindi programming in the world, with an extensive library housing over 260k+ hours of television content with more than a billion users across 173 countries. ZEE Entertainment operates 41 domestic channels and 38 international channels. Within three years of its launch, ZEE Entertainment's **ZEE5**, has become one of the biggest OTT platforms in India with approximately 65.9 million monthly active users, 5.4 million daily active users, and more than 500 million video views per month.

Zee TV India moved their SQL Production workload from Azure SQL to Amazon RDS for SQL Server in CY 2021. The core reason they moved is they were not happy with Azure Services and the scale of workload they had they felt AWS in general is a better platform. Hence since they were planning to move Application workload, they were also looking for options to move database workload as well.

Customer Type: Utility/Energy

Customer Location: Singapore

Customer Name: [Top Electricity Supplier and Retailer in Singapore | Tuas Power \(savewithtuas.com\)](#)

A customer in Singapore was planning a database migration to the cloud as part of their modernization initiative. The customer is accustomed to Azure/Microsoft platform with their strong relationship from the past. They were performing a proof-of-concept on Azure to test

out some of their success criteria as part of their migration plan. We approached the customer presenting the capabilities and options of SQL Server on AWS. Customer has agreed to do the same POC on us.

Customer has an on-premise SQL Server 2016 Enterprise Edition with Always-On Availability Group setup. The instance is hosting 1 database of 1.3TB which stores data from power meters of their customers. Data growth is expected to be about 2GB per day. It's a .NET application on the front-end and has an RPO of 4 hours and an RTO of 6 hours.

Customer has highlighted the below challenges:

1. Database backup takes a long time to complete due to the size of their database
2. Storage capacity due to expected growth
3. SQL Agent job that processes the data takes time to complete

Customer would like to see the 3 challenges mentioned above addressed when moving to the cloud. They would like to leverage a fully managed service to reduce the heavy-lifting work for their DBAs and finally a better performance to cost ratio

With these challenges and requirements in mind, we started the POC with Amazon RDS for SQL Server using db.r5b.4xlarge instance type with 3,000 PIOPS. Customer would be executing a stored procedure which will perform about 60% of data analysis of their database. The workload has 60% reads and 40% writes. Once the first test run has been completed, we tested a second test leveraging r5b.2xlarge instance type.

The results as follows:

RDS SQL Server (r5b.4xlarge)	RDS SQL Server (r5b.2xlarge)	Azure SQL MI (8 Core, General Purpose Gen5, 3TB storage)	On-Premise DEV (6 vCPU, 20GB, HDD)	On-Premise PRDO (10vCPU, 48GB, SSD)
18 mins	36 mins	145 mins	132 mins	540 mins

The results above showed a significant performance advantage running their test on RDS. Customer was pleased with the results achieved. On the first week of Jan 2023, customer has awarded the RFP to a AWS Partner to assist them in their migration.

RMS Cloud

Customer Type: ISV

RMS Cloud Overcomes “Noisy Neighbor”, Improving Database Performance by 30% on AWS

RMS Cloud transformed its multi-tenant architecture with Amazon RDS, taking advantage of prompt support from AWS and AWS Partner Slalom to achieve higher availability and eliminate downtime for customers.

[RMS Cloud](#) is an all-in-one property management platform, serving more than 7,000 hospitality properties in over 70 countries. The company recently migrated its database architecture to AWS to obtain better technical support and eliminate downtime for customers.

859 RMS Cloud is using [Amazon RDS for SQL Server](#), [Amazon EKS](#) for Kubernetes workloads, and
860 [Amazon EFS](#) for storage. Since migration, the company has achieved 20–30 percent better
861 performance and eliminated issues with downtime that often impacted its multi-tenant
862 architecture.

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Main Story Content

Opportunity | Lacking Support During an 8-Hour Outage

For more than 40 years, [RMS Cloud](#) has provided an all-in-one property management platform to hospitality businesses in Australia and internationally. In fact, more than 7,000 properties in 70 countries currently use RMS software. RMS Cloud has been in a process of digital transformation, constantly looking for new ways technology can deliver cost-efficient, resilient solutions at scale. Since 2011, the company has used a multi-tenant software model on the cloud.

As data volumes grew, the software provider faced challenges with its cloud-based database management service, often experiencing downtime during traffic surges. Complications with one customer would often impact other tenants sharing the same database instance, leading to widespread service downtime. To address potential client complaints, RMS Cloud resorted to assigning customers to individual single-tenant instances—a costly solution that undermined the benefits of a multi-tenant model.

In January 2023—one of the busiest months for the Australian hospitality industry—the company’s cloud provider suffered a significant outage that resulted in nearly 8 hours of downtime overnight. RMS Cloud is rapidly growing in Asia and Europe, and the outage did not bode well for this expansion. More importantly, engineers had little to no support from the provider during the event. RMS Cloud soon decided to switch to a provider that could offer higher resilience and also address the [“noisy neighbor” issue](#) with its multi-tenant database model.

Pull Quote

4 lines max.

“We’re not babysitting the SQL servers anymore, which helps us really focus on our application development rather than database management.”

- Andrew Buttigieg, Technical Services Manager at RMS Cloud

Solution | Achieving Seamless Migration with Higher Availability and Security

RMS Cloud found the support it was seeking in Amazon Web Services (AWS). The company’s AWS account team connected it with [AWS Partner Slalom](#) to build a modern cloud strategy. What began as a straightforward migration grew to encompass higher availability, better performance, and lower database cost on AWS.

Slalom initiated a comprehensive cost and technology analysis using the [AWS Well Architected](#) framework, spending nearly three months designing and stress-testing new, customized architecture before migrating any production workloads. The assessment was fully funded by AWS as part of the [AWS Migration Acceleration Program](#) (MAP), which ensured technical acceleration and commercial benefits throughout the migration.

Andrew Buttigieg, technical services manager at RMS Cloud, says, “The transition was very smooth. Slalom and AWS have both been very responsive throughout the process, with response times from a few minutes to an hour at most.”

As part of the [AWS Enterprise Support](#) plan, RMS Cloud benefits from AWS [Infrastructure Event Management](#) (IEM) during migration waves. By February 2024, about six months after the first wave, the company will have migrated all 4,300 databases—more than 2 TB of data—to [Amazon RDS for SQL Server](#). RMS Cloud also seamlessly moved its Kubernetes workloads to [Amazon Elastic Kubernetes Service](#) (Amazon EKS) and is using [Amazon Elastic File System](#) (Amazon EFS) for serverless storage.

Along with the migration, RMS Cloud has implemented additional security features such as [Amazon Inspector](#) to detect software vulnerabilities and centralize security alerts. Data is encrypted both in transit and at rest, and backups occur automatically within AWS Regions in Australia. “We can avoid security drift thanks to the thorough architecture review we’ve done prior to migration,” adds Buttigieg. “With the modernization of our database, we’ve built in additional encryption for PII [personally identifiable information].”

Outcome | Resolving Downtime and Addressing the "Noisy Neighbor"

By migrating to AWS, RMS Cloud achieved a performance gain of 20–30 percent for its databases. Amazon RDS offers a lower CPU utilization rate compared to RMS Cloud’s previous provider, which means the company can run smaller machine sizes for a similar customer pool and save on cost per customer.

The “noisy neighbor” issue that was causing downtime has also been resolved. Despite facing similar spikes in demand, these are now swiftly managed on Amazon RDS, preventing disruptions from spreading to other customers on the same server. Today, RMS Cloud observes a maximum delay of 15–20 seconds, in contrast to the 5- to 6-minute delays experienced with the previous cloud database. “We’re not babysitting the SQL servers anymore, which helps us really focus on our application development rather than database management,” Buttigieg adds.

Additionally, RMS Cloud uses service-level granular reporting via [AWS Cost Explorer](#), empowering the business to visualize, comprehend, and manage AWS costs and usage effectively. This meticulous monitoring of costs and consumption has facilitated a budget-controlled migration and established a solid foundation for future modernization initiatives.

Most importantly, RMS Cloud feels confident in the support they receive from AWS. Peter Buttigieg, managing director at RMS Cloud, says, “Our business requires 24/7 uptime. While challenges are inevitable, the ability to quickly connect with experts who understand our needs and respond rapidly has been a game changer for us.” Next on its cloud roadmap, RMS Cloud plans to implement [Amazon Aurora](#) for PostgreSQL.

“

We're not babysitting SQL servers anymore, which helps us really focus on our application development rather than database management.

Andrew Buttigieg
Technical Services Manager at
RMS Cloud

”



INDUSTRY
Software & Internet
COUNTRY
Australia

ABOUT

RMS Cloud delivers a scalable, all-in-one property management platform used by hospitality businesses in more than 70 countries. Launched in Australia 40 years ago, the company currently serves over 7,000 unique properties.

RMS Cloud Overcomes “Noisy Neighbor”, Improving Database Performance by 30% on AWS



Business Needs

RMS Cloud faced challenges with its cloud-based database management service, often experiencing downtime during traffic surges. It needed a cloud provider that could offer higher resilience and also address the “noisy neighbor” issue with its multi-tenant database model.



Solutions

- Amazon RDS for SQL Server
- Amazon EKS
- Amazon EFS
- Amazon Inspector
- IEM
- AWS Cost Explorer



Outcome

- **20–30%** – better CPU utilization for database instances
- **Robust data protection** – Prevents security drift
- **24/7 support** – from local AWS teams
- **Zero downtime** – Improved resilience in multi-tenant model

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Summary

Overall, Amazon RDS for SQL Server is a compelling option that can cover the breadth of configurations offered by Azure SQL. Additionally, RDS offers the security and resilience of AWS and integration with a breadth of services providing the greatest depth in Analytics, AI, Machine Learning, Compute, and more.

Amazon RDS for SQL Server is a managed database which provides customers with:

- A choice of higher performance hardware and more infrastructure configurability than Azure SQL. This allows customers to right-size their workloads, decreasing operating and licensing costs. Higher performance configurations include faster CPU clock speeds, larger instance sizes, higher IOPS, and more storage throughput.
- License/Cost Flexibility & Elasticity. With RDS’s License Included model, customers pay for what they use. When the database is not running, customers do not pay for the license. It is a convenient way to migrate without committing to a 3-year Microsoft Enterprise Agreement (EA) + Software Assurance expense. Azure SQL does offer a pay-as-you-go licensing model as well, but using this model on RDS could be preferable to customers managing their own licenses manually.
- Amazon RDS Custom retains the managed benefits but also offers access to the underlying host operating system. For more details please refer to [Azure Compete Wisdom Page](#).

Frequently Asked Questions (FAQ):

Can AWS support SQL Server workloads?

Yes, Amazon RDS for SQL Server is a mature, fully-managed database service with SQL Server

engine compatibility. Customers have been running SQL Server on RDS since 2012 and can choose from a range of powerful compute options, benefit from resilient underlying infrastructure, and integrate with the rest of the AWS data ecosystem.

What are the strengths of Amazon RDS?

RDS for SQL Server is a service built on strong AWS infrastructure. Customers have a wide range of compute instances to choose from, including memory-optimized options which can improve performance. All RDS databases utilize EBS for data storage and retrieval, giving users single-millisecond latencies for many workloads. And building on the resilient AWS global infrastructure gives customers access to 3 or more Availability Zones in every region, making it simple to plan for High Availability.

What should I do if my customer is running Elastic Pools, Hyperscale, or Serverless on Azure?

Elastic Pools, Hyperscale, and Serverless are three deployment options which do not have direct comparisons in RDS for SQL Server. However, RDS can still support your customer's workloads. Before proposing a solution, gather information about your customer's use case and requirements (how many databases are they running, how large are these databases, how do they scale, and how are they connected to other applications). Then, you can design for those requirements on RDS or reach out to a Specialist via the SpecReq tool for support.

What are the licensing options on RDS for SQL Server?

RDS for SQL Server only offers a License-Included purchasing model. This means that AWS manages the purchase and compliance of licenses, and customers pay an hourly rate for the licenses while their databases are running. The model removes complexity from managing licenses and gives customers the option to save on cost by paying only for what they use. However, if customers have existing investment in SQL Server licenses, they may want to consider BYOL for SQL Server on EC2.

What is Azure Hybrid Benefit, and how should I respond to it?

Azure Hybrid Benefit provides customers with the ability to bring existing Microsoft SQL Server licenses (with active Software Assurance) to the cloud, offsetting the license costs for Microsoft Azure managed SQL offerings. First, be sure that the customer actually qualifies for Hybrid Benefit. If they do, then RDS may still make sense depending on the customer workload. You could also recommend Bring-Your-Own-License for SQL Server on EC2 as a cost reducing option.

Are there programs available for reducing cost of RDS for SQL Server?

There are a variety of programs available for reducing costs of a migration. The [Migration Acceleration Program](#) is an excellent place to start and offers general benefits as well as additional discounts for migrating SQL Server workloads.

Azure SQL Database provides Auto Tuning feature however RDS SQL Server although it is a managed offering doesn't have similar capabilities. Any timeline we can expect this feature in RDS?

Overall RDS has multiple Cloud native tools to provide a comprehensive monitoring of SQL Server. However, we do not provide cost breakdown of individual databases and Auto Tuning capabilities of SQL Server Databases similar to Azure SQL. However automatic tuning feature of Azure SQL Database is pretty basic and recommendations are not too aligned during complex TSQL query optimization. In RDS SQL Server rebuild index is generally not required due to the way EBS storage is designed. You can still do rebuild index and it can be automated through SQL

Agent jobs. Regarding missing index or unused index recommendation, this still can be worked out by providing custom T SQL scripts which can be scheduled once through SQL Agent job. Going back to other tuning opportunities, with SQL Server 2017 and beyond on RDS, it by default provides self-tuning capabilities in the database engine itself. And this is what even Auto tuning feature in Azure uses.

Is database storage tightly coupled to compute resources or independently scalable?

On AWS, database storage is decoupled and able to scale up and down regardless of the database compute resources. While Microsoft Azure vCore pricing model does allow some flexibility, storage and compute resources are more tightly coupled. For example, in order to reach the maximum 4TB database size on Microsoft Azure General Purpose tier, customers must provision at least 24 vCores. This can lead to over-provisioning and unnecessary costs.

How do the High Availability configurations differ on each cloud?

On Microsoft Azure Business Critical tier, customers receive a primary instance, 2 high availability replicas, and one read replica. On AWS, customers have more flexibility in their HA configurations, with the option to have a primary instance, a replica in multi-AZ configuration, and up to 5 read replicas to reduce load on the primary instance.

What instance types are available?

Both AWS and Azure offer a variety of instance types with different processors, Memory-to-CPU ratios, and overall performance characteristics. For a full list, see the respective pages for [RDS](#), [Azure SQL Database](#), and [Azure SQL Managed Instance](#).

What is the available Memory-to-CPU ratios?

AWS offers a wide range of instance types with a variety of Memory-to-CPU ratios. Most of the general-purpose instances offer ratios of around 4:1, while customers can use memory-optimized instances with ratios as high as 30.5:1. These higher-memory instances can help improve efficiency and reduce cost for many memory-bound SQL workloads. On Azure, the General-Purpose instances offer Memory-to-CPU ratios of around 5:1, and they offer memory-optimized instances with ratios of around 29:1.

[Amazon RDS Instance Types | Cloud Relational Database | Amazon Web Services Pricing - Azure SQL Database Single Database | Microsoft Azure Pricing – Azure SQL Managed Instance Single Instance | Microsoft Azure](#)

APPENDIX 1:

The following regions currently support availability zones: For more details click [here](#).

	Americas	Europe	Middle East	Africa	Asia Pacific
1	Brazil South	France Central	Qatar Central	South Africa North	Australia East
2	Canada Central	Germany West Central	UAE North		Central India
3	Central US	North Europe			Japan East
4	East US	Norway East			Korea Central
5	East US 2	UK South			Southeast Asia

6	South Central US	West Europe			East Asia
7	US Gov Virginia	Sweden Central			China North 3
8	West US 2	Switzerland North			
9	West US 3				

Azure services with availability zone support

Azure services that support availability zones, including zonal and zone-redundant offerings, are continually expanding.

Three types of Azure services support availability zones: *zonal*, *zone-redundant*, and *always-available* services. You can combine all three of these approaches to architecture when you design your reliability strategy.

- **Zonal services:** A resource can be deployed to a specific, self-selected availability zone to achieve more stringent latency or performance requirements. Resiliency is self-architected by replicating applications and data to one or more zones within the region. Resources can be pinned to a specific zone. For example, virtual machines, managed disks, or standard IP addresses can be pinned to a specific zone, which allows for increased resiliency by having one or more instances of resources spread across zones.
- **Zone-redundant services:** Resources are replicated or distributed across zones automatically. For example, zone-redundant services replicate the data across three zones so that a failure in one zone doesn't affect the high availability of the data.
- **Always-available services:** Always available across all Azure geographies and are resilient to zone-wide outages and region-wide outages. For a complete list of always-available services, also called non-regional services, in Azure, see [Products available by region](#).

For more information on older-generation virtual machines, see [Previous generations of virtual machine sizes](#).

Appendix 2

<https://wisdom.corp.amazon.com/Pages/Cloud-Intelligence-Microsoft-Azure-SQL.aspx>

Other Call Outs

Azure SQL Elastic Pool to RDS SQL Server Right Sizing & Modelling

To calculate the TCO for moving Azure SQL Elastic Pool databases to RDS for SQL Server, use the following Cost Modelling and Right Sizing Tool.

<https://amazon.awsapps.com/workdocs/index.html#/document/f0929bff67a93328dba86005c7deff97004ac0f021c63039f4d5bf957e312569>

Customers can create and manage SQL Server Agent Jobs with RDS. Azure SQL Database doesn't have SQL Server Agent, which is used to schedule jobs and send success/failure

1081 notifications. Azure SQL Managed Instance has SQL Server Agent and can be used to schedule
1082 jobs just like with on-premises SQL Server

1083

1084 Customers can run three-part (cross-database) and four-part name (Linked server) queries on
1085 RDS. Customers can't access the tables in different SQL Databases in Azure on the same Azure
1086 SQL Server using three-part names. Four-part names
1087 (ServerName.DatabaseName.SchemaName.TableName) aren't allowed at all. Multi-part names
1088 and cross-database queries are supported on SQL Managed Instance. Customers will need to
1089 refactor their code having three-part or four-part name queries before migrating to Azure SQL
1090 database.

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1092

1093 **The End**

1094